

NORTH DEERING NEIGHBORHOOD ANALYSIS

Introduction

In MONTH YEAR, a group of concerned citizens contacted the Maine Cancer Registry with concerns regarding the number of persons diagnosed with cancer in Portland's North Deering neighborhood. The following information is based on analysis completed to assist with the investigation into the number of incident cancer cases in the North Deering neighborhood. Of particular interest was the suspected number of incident cases found in the youth/young adult population. The intent of the analysis was to see whether the number of incident cancer cases in the North Deering neighborhood differed from what you would expect to find in a Maine population.

Methods

Demographic information on the cases identified by the concerned citizens was procured and cross checked with the Maine Cancer Registry tumor database (IMPAC) for accuracy. There were 3,398 incident tumor cases identified by the Maine Cancer Registry for Portland during the time period of 1995-2004 using IMPAC. Cumberland County and Maine State cases counts were retrieved using SEER Stat software.

Portland case addresses were geocoded according to their latitudinal and longitudinal coordinates using XXX software. Of the 3,398 tumor cases, 136 (4.0%) were not geocoded due to either a missing address or post office box listing and excluded from the analysis. Using GIS software, cases were then categorized and coded into one of the Portland Census tracts.¹

Census 2000 data were used to estimate the population of North Deering and Portland.² Post-2000 and inter-censal estimates were used for Cumberland County and Maine State estimates (1995-1999 and 2000-2004, respectively).

Cases were stratified by geography, age at diagnosis, and selected major cancer site. Four strata were constructed for the analysis; North Deering, Portland (excluding North Deering hereafter referred to as just "Portland"), Cumberland County, and Maine State. Two census tracts (23005002101 and 23005002200) were identified as belonging to the North Deering neighborhood. The remaining seventeen census tracts were identified as belonging to the strata named Portland. Age groups used in the analysis were 0-19 years, 20-44 years, 45-64 years, 65-74 years, 75-84 years, and 85 plus years. Cancers were categorized according to the SEER incident cancer site recode definitions.³

¹ Portland Census tracts: 23005000100, 23005000200, 23005000300, 23005000500, 23005000600, 23005001000, 23005001100, 23005001200, 23005001300, 23005001500, 23005001700, 23005001800, 23005001900, 23005002001, 23005002002, 23005002101, 23005002102, 23005002200, 23005002300.

² Census Bureau. American Factfinder Census 2000 dataset. Available URL: <http://www.census.gov>.

³ National Cancer Institute (NCI), Surveillance Epidemiology and End Results (SEER) Incidence site recode. Available at: <http://seer.cancer.gov/siterecode/index.html>

In the youngest age group, 0-19 years, cancers were grouped and analyzed by the following categories; all cancers, lymphocytic leukemia, acute lymphocytic leukemia, myeloid and monocytic leukemia, other leukemia, Hodgkin lymphoma, Non-Hodgkin lymphoma, and brain cancer. All of the other age groups had incident cancer cases analyzed according to the following groups; all cancers, lung, colon, breast (female only), and prostate cancer. Three additional cancer sites were analyzed for the 20-44 year old age group; melanoma, thyroid, and kidney.

Rates and accompanying 95% confidence intervals were calculated using SAS software for the North Deering and Portland analysis⁴. The expected number of cases was calculated for North Deering based on each of the three remaining strata, i.e. Portland, Cumberland County and Maine State.

Results

Between 1995 and 2004 there were 10 cases of incident malignant cancers (insitu included for brain cancers) found in persons 0-19 years of age in the North Deering neighborhood, 19 cases in Portland, 139 cases in Cumberland County, and 609 cases in total for Maine (Table 1). Of the ten cases found in North Deering, eight are accounted for in the individual incident site categories presented next. One case each of respiratory and endocrine site cancers were not included in the individual site analysis (data not shown).

In persons aged 0-19 years there were no differences in the observed number of cases when using Cumberland County or Maine as the basis for acute lymphocytic, and brain cancer (Tables 2-3). North Deering experienced higher than the expected number of cases of myeloid and monocytic leukemia (17.0 per 100,000), and Hodgkin lymphoma (17.0 per 100,000) compared to Cumberland County (1.2 and 2.3 per 100,000, respectively) and Maine (1.0 and 1.6 per 100,000, respectively)(Tables 4-5). There were no cases of other leukemia or Non-Hodgkin lymphoma reported in North Deering during the same time period (1995-2004)(Table 6).

For the same time period, there were 53 cases of incident malignant cancers found in persons 20-44 years of age in North Deering neighborhood, 198 cases in Portland, 1,061 cases in Cumberland County, and 5,081 in total for Maine (Table 7). Of the 53 cases found in North Deering, 31 are accounted for in the individual incident site categories presented. Of the twenty-two incident cancers not compared in the following analysis, 6 were lymphoma, 3 cervix or ovary, 2 oral cavity, 3 digestive, 1 respiratory, 1 brain, 2 testis, 1 invalid (data not shown)(*I know I'm missing 3 cancers in the group*).

In persons aged 20-44 years there were no differences in the observed number of cases when using Portland, Cumberland County or Maine as the basis for lung, colon, breast, and prostate cancers (Tables 8-11). North Deering experienced higher than the expected number of cases of melanoma (26.2 per 100,000) compared to Portland, Cumberland County, and Maine (7.2, 10.8, and 12.4 per 100,000, respectively)(Table 12). North

⁴ SAS Institute, Inc. (2002-2003). *SAS Release 9.1.3*. Cary, NC.

Deering experienced a higher number of thyroid and kidney cancers (17.5 and 8.7 per 100,000, respectively) compared to Portland (6.0 and 1.8 per 100,000, respectively)(Tables 13-14). There were no observed differences in the expected number of thyroid or kidney cancer in North Deering compared to Cumberland County or Maine.

For the same time period in persons aged 45-64 years, there were 162 cases of incident malignant cancers in North Deering neighborhood, 705 cases in Portland, 4,386 cases in Cumberland County, and 23,016 in total for Maine (Table 15). Of the 162 incident cases in North Deering, 94 are accounted for in the site specific analysis.

There were no differences in the observed number of incident cases of lung, colon, and prostate cancer in North Deering compared to Portland, Cumberland County, or Maine (Tables 16-17, 19). The number of breast cancer cases was higher than expected in North Deering (340.4 per 100,000) compared to Portland (251.1 per 100,000), Cumberland County (279.9 per 100,000), and Maine (262.0 per 100,000)(Table 18).

In persons aged 65-84 years there were 328 cases of incident malignant cancers in the North Deering neighborhood, 1,212 cases in Portland, 6,876 cases in Cumberland County, and 38,593 cases in Maine (Table 20). Of the 328 incident cases in North Deering, 180 are accounted for in the site specific analysis.

No differences were observed in the number of incident cases of colon cancer in North Deering compared to Portland, Cumberland County, and Maine (Table 22). North Deering had lower than expected numbers of lung cancer (384.1 per 100,000) compared to Portland (469.6 per 100,000), Cumberland County (427.7 per 100,000), and Maine (443.4 per 100,000)(Table 21). The number of incident cases of prostate cancer were also lower in North Deering (705.7 per 100,000) compared to Cumberland County (806.4 per 100,000) and Maine (980.6 per 100,000)(Table 24). The number of newly identified breast cancers in North Deering (456.8 per 100,000) were higher when compared to Portland (371.0 per 100,000)(Table 23).

In persons aged 85years or more, there were 89 cases of incident malignant cancers in the North Deering neighborhood, 248 cases in Portland, 1,171 cases in Cumberland County, and 5,635 cases in Maine from 1995 to 2004 (Table 25). Of the 89 incident cases identified, 42 are accounted for in the site specific analysis.

There were no differences in the observed number of cases of breast cancer in North Deering compared to Portland, Cumberland County, or Maine (Table 28). There were fewer cases of colon cancer in North Deering (288.2 per 100,000) compared to Portland (456.2 per 100,000)(Table 27). In North Deering there were higher than expected numbers of lung cancer (432.2 per 100,000) compared to Portland (256.7 per 100,000), Cumberland County (320.2 per 100,000), and Maine (306.9 per 100,000)(Table 26). Prostate cancer was higher in North Deering (1,020.4 per 100,000) compared to Portland (393.0 per 100,000)(Table 29).

Limitations

There are a few limitations for this analysis. First, the small number of cases identified in the North Deering neighborhood limits the precision of our estimates. Additionally, we identified that of the 136 cases not geocoded and excluded from the analysis 132 were malignant tumors, therefore we do not know if they were missing at random across both site and geography (North Deering versus Portland)(data not shown). In the analysis we identified a few cancer sites that only differed between the North Deering neighborhood and Portland such as, prostate cancer in 65-84 and 85 plus years of age, breast cancer 65-84 years of age, thyroid and kidney cancer in 20-44 year olds. Each of these may have been affected by the missing cases. For example, in the missing cases we identified 10 prostate cancers in men 45-64 years of age, if only 1 of those cases is located in North Deering then the difference in observed numbers between North Deering and Portland goes away (data not shown). This demonstrates the instability of working with small numbers and the effect even small shifts in unknowns may have on the ultimate analysis.

Finally, the intent of this report is to present findings of base analysis only. The report is not a complete description of the cancer types and their relationship to or identification of potential environmental exposures.